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CS 523 Rust Programming
Course Project Proposal
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Rumbo: Camp for Free in Oregon

What if I told you there were still pockets of wilderness you can camp at without giving a single dollar to the man? For many, spending the night at a campground is the perfect balance of civilization and nature, getting to see trees and wildlife while still having access to modern amenities like running water and restrooms. For others, a desire remains to venture further still, take the next step away from the warm shallows and explore deeper into the natural world. Dispersed camping is one path towards exploring further into the outdoors: it describes camping in large areas that have been designated as public land, legal to camp in for ~14 days at a time but with no campground or amenities provided. This land is your birthright as an American citizen, but the responsibility falls on you to figure out when, where and how to explore it responsibly.

Part of the sport of dispersed camping is finding spots that have the potential to make the best campsites. In most states including Oregon, this means combing through maps and triangulating what spots are accessible by your mode of transportation, what could have the potential to have a great view of the sunrise in the morning, where you might find solitude while remaining close to home. Everyone deserves the tools to find great spots by themselves, and right now I find the tools to do so inadequate. The current best free options is the BLM's own official map, but this map lacks the critical ability to pinpoint Latitude/Longitude on locations of interest, as well as suffering from a poor contrast color scheme and the inability to switch base maps.

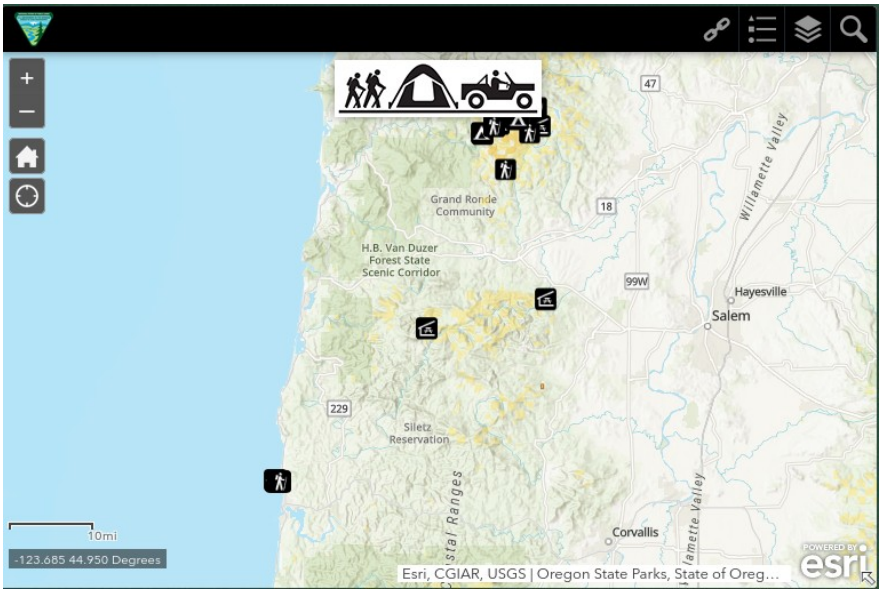


Figure 1: Poor contrast and features for saving points of interest

Other options exist from purpose built camping/hunting applications like onX Offroad or Gaia GPS, but exist behind paywalls or login requirements. I think we can do better considering the underlying geographic data is free to download or query from the Bureau of Land Management itself.

This project will be called Rumbo, which means heading or direction in Spanish. I aim to build the foundation of a simple and responsive website that allows users to view BLM land they are able to camp on. Strech/further goals include allowing users to pinpoint Lat/Long of points of interest, add road segments to help users have better context of how they can access their points of interest, and allowing users to save and export multiple points of interest into other common mapping applications like Google Maps.

This begins with building and Axum server in Rust that queries the BLM's mapserver for BLM owned areas, then processing and applying them to a static base map. A base map can be sourced from OpenStreetMap, and having the option to switch from a cartographic basemap to a satellite base map is another feature I'd like to implement. While I have experience working with web maps, using Rust to help build the website will be a new experience for me, and I expect that to be the most challenging part of implementing this project.

I'm really excited for the opportunity to work on this project as it's something I've had in mind for a while, I appreciate being granted the theme opt out to do it and my goal is to give myself a baseline for a long term project I can continue to expand and work on after this class is completed.